Be a Word Architect

Vocabulary Words

autobiographical
autobiography
autograph
biographical
biography
graphic
graphics
graphite
monograph
oceanography
paragraph
phonograph
photograph
photographic
photography

Word Learning Tip!

Some words belong to word families. A word family is all the words that share the same main word or root and have related meanings. *Graph* is the main word in the *graph* word family. It comes from an ancient Greek word meaning "to draw or write." Look at all the words you can build from the word part *graph*.

Vocabulary Building Strategy

Use Word Families When you read an unfamiliar word that contains *graph*, you can infer that it has something to do with writing, drawing, or communications. Add the meaning of the main word or root in a word family to other word parts, such as prefixes and suffixes, to learn the meaning of the new words that are in that same word family.

Learn More!

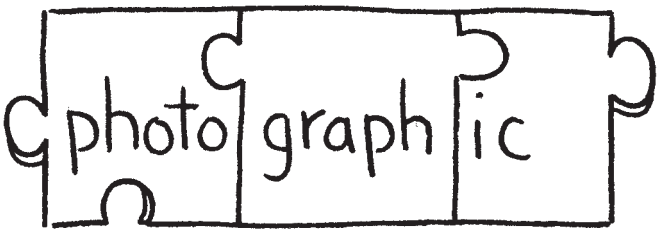
You can use *graph* to build new words by adding prefixes, suffixes, roots, and even whole words. For example, photo + graph + ic = photographic. Here are some prefixes, suffixes, and roots you can add to *graph* to form words.

Prefix	Meaning
auto-	self
bio-	life
mono-	one
para-	beside; almost

Suffix	Meaning
-ic, -ical	relating to
-ics	scientific or social
-ite	mineral or rock
-y	activity

Root	Meaning
phono	sound
photo	light

Many English words for inventions having to do with communication have been made up using *graph*.



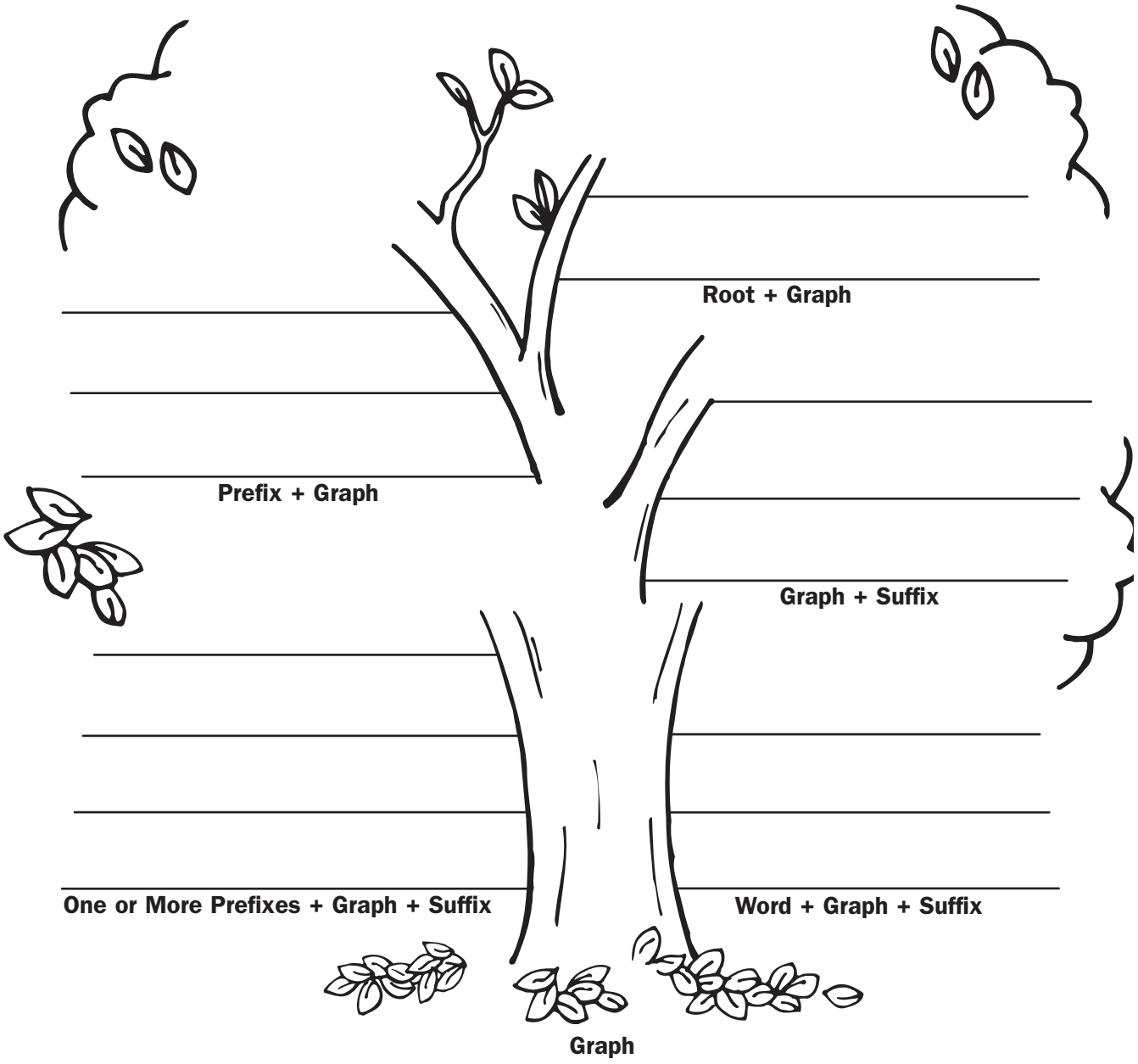
Collect Autographs Some people have framed autographs or signatures from people they admire. Write a paragraph describing what autographs you might like to collect. Give reasons to support your choices. Use as many "graph" words as you can.

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Be a Word Architect

autobiographical	biographical	graphics	oceanography	photograph
autobiography	biography	graphite	paragraph	photographic
autograph	graphic	monograph	phonograph	photography

Directions Look at each vocabulary word. Try to determine how it was formed. Then write it on the correct branch of the tree below. Circle *graph* in each word. (Hint: The word *oceanography* contains an *o* before *graph* to make it easier to pronounce.)



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Connect Words and Meanings

autobiographical	biographical	graphics	oceanography	photograph
autobiography	biography	graphite	paragraph	photographic
autograph	graphic	monograph	phonograph	photography

Directions Read each definition below. Then complete the sentence that follows it.

1. **oceanography** the science dealing with oceans and the plants and animals that live in them

   In a book on **oceanography**, I would find information about _____

   _____

2. **autograph** a person's own signature; to write your name or signature on something

   I would like an **autograph** from _____

3. **graphics** images such as drawings, maps, and graphs, especially those made using technological devices such as computers

   When I use a textbook, the **graphics** help me because _____

   _____

4. **phonograph** a machine that plays disk-shaped records

   Instead of a **phonograph**, most people today have _____

   _____

5. **paragraph** a series of sentences that develop one main idea

   When I write, I begin a new **paragraph** by _____

   _____

6. **autobiography** the life story of a person written by that person

   An **autobiography** that I would like to read would be written by _____

   _____

7. **photograph** a picture made by the action of light upon a surface; a picture taken by a camera

   I treasure my **photograph** of _____

8. **monograph** a short book or long article on a single, limited subject

   If I were researching space travel, I might read a **monograph** on the topic of _____

   _____

Continued on next page

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Connect More Words and Meanings

autobiographical	**biographical**	**graphics**	**oceanography**	**photograph**
autobiography	**biography**	**graphite**	**paragraph**	**photographic**
autograph	**graphic**	**monograph**	**phonograph**	**photography**

Directions Continue this activity. Read each definition below. Then complete the sentence.

9. **graphite** a soft, black mineral used as lead in pencils and also used in paints and coatings
   The company that supplies **graphite** increased production when it received a large order from a school for _____

10. **biography** the life story of a person written by someone other than that person
    I would like to read a **biography** of _____

11. **photography** the making of pictures by exposing film in a camera to light
    If I wanted to study **photography**, I would need _____

12. **autobiographical** having to do with your own life story
    I once told an **autobiographical** story about _____

13. **biographical** having to do with someone else's life
    A **biographical** detail I would like to know about my favorite pop star is _____
    _____

14. **graphic** having to do with art and design or handwriting; very realistic
    A **graphic** novel is fun to read because _____

15. **photographic** having to do with recalling from memory the exact order in which a long list of names was read; having to do with photography
    A person's memory is **photographic** when _____

Have a Biographical Chat In a small group, make a list of biographies and autobiographies you have read. Then choose one and write a paragraph telling why you liked it. Use at least four *graph* words in your paragraph.

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Learn Words in Context

autobiographical	biographical	graphics	oceanography	photograph
autobiography	biography	graphite	paragraph	photographic
autograph	graphic	monograph	phonograph	photography

Interview With a Writer



Interviewer: Good morning, Dr. Lucille Luce. You are known as the world's busiest writer, aren't you?

Dr. Luce: Yes, that's right. So far, I have written twelve **biographies** of famous people and dozens of **monographs** on scientific subjects. And I'm planning to write a full-scale, completely truthful **autobiography** in which I tell how my **photographic** memory has helped me . . . as soon as I can find the time. I have written an introduction to the underwater science of **oceanography** for young people, and I have also written a manual on how to take and develop better **photographs**. **Photography** is one of my hobbies, you see, and I want to share my knowledge of cameras, lenses, developing, printing, and everything else that has to do with taking good pictures.

Interviewer: Gee, that's wonderful.

Dr. Luce: Yes, it is. I might add that I'm currently researching a book on Thomas Edison's invention of the **phonograph**, which was of course the greatest step forward in recording sound. I'm planning for that book to have excellent **graphics**, including old photos of Edison at work, and diagrams of how sound is recorded. The **graphic** element is important to me as an author, because people learn from pictures as well as from words. For example, in an **autobiographical** work, I would include photos of myself from throughout my life. In planning a **biographical** work, I always search carefully for pictures of my subject and of where he or she lived.

Interviewer: And yet the words are the most important thing for you as a writer, aren't they?

Dr. Luce: Of course. I use up the **graphite** in pencil after pencil, revising and revising until I get it just right. When I revise, I change single words, phrases, sentences, **paragraphs**, and even entire chapters.

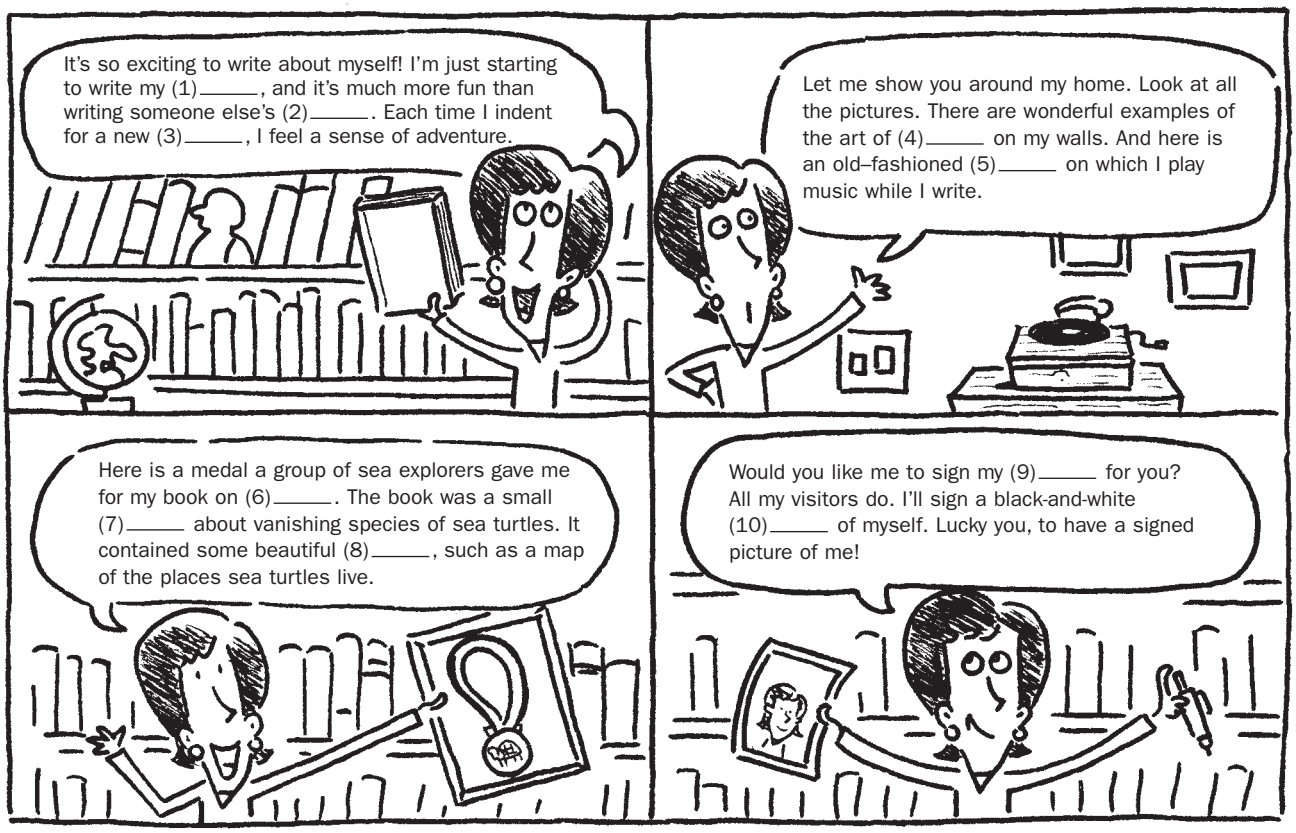
Interviewer: That's so inspiring! May I have your **autograph** for my collection of famous people's handwriting?

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Use Words in Context

autobiographical	**biographical**	**graphics**	**oceanography**	**photograph**
autobiography	**biography**	**graphite**	**paragraph**	**photographic**
autograph	**graphic**	**monograph**	**phonograph**	**photography**

Directions Read the comic strip below. Fill in the blanks below the comic strip with the correct vocabulary words.



1. _____ **5.** _____ **9.** _____

2. _____ **6.** _____ **10.** _____

3. _____ **7.** _____

4. _____ **8.** _____

Extend the Comic Strip Add a frame to the comic strip. Use at least one vocabulary word in the new frame.

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Review and Extend

autobiographical	**biographical**	**graphics**	**oceanography**	**photograph**
autobiography	**biography**	**graphite**	**paragraph**	**photographic**
autograph	**graphic**	**monograph**	**phonograph**	**photography**

Directions The suffix *-er* means "a person who." Add the suffix *-er* to a vocabulary word to answer each question below. If the vocabulary word contains a suffix, drop the suffix before you add *-er*.

1. A person who takes pictures with a camera is a(n) _____
2. A scientist who studies the oceans is a(n) _____
3. Someone who writes about another person's life is a(n) _____
4. Someone who writes articles about a single subject is a(n) _____

Directions Use your knowledge of word parts to find the meaning of each word below. Check your answers against the dictionary.

New Words

geographer **graphology** **seismograph**

5. The prefix *seismo-* means "earthquake." What do you think a **seismograph** is?

_____

6. The prefix *geo-* means "earth." What do you think a **geographer** does?

_____

7. The suffix *-ology* means "study of or science of." What do you think **graphology** is?

_____

8. The suffix *-ist* means "a person who." What do you think a **graphologist** does?

_____

Write About It Write about the kind of "grapher" you might like to be when you grow up. Why? What kind of "graphy" would you most like to do? Why? Read your ideas to a partner.

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Check Your Mastery

Directions Read each item below. Write the word in the blank that best completes each sentence.

1. Author Peter Proffitt is writing a new _____ (*biography, autobiography, photograph*) of Caesar Chavez.

2. When I am old, I hope to write my _____ (*autobiography, oceanography, paragraph*) so that others will learn from my experiences.

3. A person who can recall the exact order of a long list of names is said to have a _____ (*monograph, oceanography, photographic*) memory.

4. She spends all her allowance on film, lenses, and camera equipment. She's very interested in _____ (*biography, photography, oceanography*).

5. The _____ (*graphite, graphics, graphic*) in this video game are colorful and clear.

6. In the days before compact discs, people played records on a _____ (*photographic, photograph, phonograph*).

7. Leela went on a cruise to Australia's Great Barrier Reef to study the science of _____ (*photography, oceanography, biography*).

8. Because _____ (*graphite, graphics, autobiography*) is black and soft, it is used in pencils.

9. She has written a small but important _____ (*phonograph, monograph, biography*) on the history of photography.

10. This diagram is a _____ (*graphic, graphite, autograph*) representation of the nation's economic growth.

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